

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8 FT 1 HT	(a)	(i)		23 units		1		1	1	
		(ii)		Time before driving = 14 hours (1) 23 units (ecf) not processed so unable to drive/should allow 23 hours (1) OR 9 unit over so not safe to drive (2) OR Won't be safe to drive until 10 pm (2)			2	2		
	(b)	(i)		Servings are of different sizes/amounts (1) so need to scale up/down the figures/ should be per 100g (1)			2	2		
		(ii)		Wheat Squares (1) No salt/highest protein /highest fibre(1)		2		2		
	(c)			BER = $32 \times 70 = 2240$ (1) EER = $8.5 \times 4 \times 70 = 2380$ (1) PER = BER (ecf) + EER (ecf) = 4 620 units (1)	1 1	1		3	3	

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					AO1	AO2	AO3	Total	Maths	Prac									
	(d)	(i)		90 kg Accept 86.7 kg		1		1											
		(ii)		As BMI doubles from 14 to 28 mass also doubles from 40 to 80 Therefore yes they are proportional. 2 marks for above. 1 mark if final 'yes' statement not included			2	2	2										
		(iii)		Subs 75/2.56 or 2.6 (1) = 29.3/29.29 or 28.8/29 (1)	1	1		2	2										
	(e)	(i)		<table border="1"><tr><td></td><td>d</td><td>d (1)</td></tr><tr><td>D</td><td>Dd</td><td>Dd</td></tr><tr><td>d (1)</td><td>dd</td><td>dd (1 ecf)</td></tr></table>		d	d (1)	D	Dd	Dd	d (1)	dd	dd (1 ecf)		3		3		
	d	d (1)																	
D	Dd	Dd																	
d (1)	dd	dd (1 ecf)																	
		(ii)		50 % (must be consistent with Punnett square) If Punnett square is blank accept 50%		1		1	1										
				Question 8 total	3	10	6	19	9										